James J. Herlan

PhD Candidate in Biology — Ecology, Evolutionary Biology, and Behavior

The Graduate Center, CUNY · New York, NY
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Education

2022–2026 **Doctor of Philosophy (PhD) Candidate**
The Graduate Center, CUNY · New York, NY
- Dissertation defense: December 2026
- NOAA EPP Earth System Sciences and Remote Sensing Scholar
- Subprogram: Ecology, Evolutionary Biology, and Behavior
- Research: Spatial ecology and co-occurrence of corals at Rapa Nui
- Keywords: species interactions, co-occurrence, dispersion · GPA: 3.8

2020–2023 **Master of Science (MS)**
College of Staten Island, CUNY · New York, NY
- NOAA EPP Earth System Sciences and Remote Sensing Scholar
- Research: Bleaching of *Porites lobata* and *Pocillopora* spp. during ENSO at Rapa Nui · GPA: 3.9

2006 **Bachelor of Science (BS)**
University of Miami · Miami, FL
Double Major: Marine Science / Biology

Research Experience

Aug–Dec 2026 **Research Associate**
North Atlantic Coast CESU · Sandy Hook, NJ
Horseshoe Cove Marsh Loss Investigation: utilize new and existing datasets to develop a process model of the site to inform recommendations of potential management actions to improve current and future marsh function. Federal Grant No. P25AC00204-00

Jun–Aug 2025 **NOAA NERTO Fellow**
James J. Howard Marine Sciences Laboratory · Sandy Hook, NJ
Correlating species occurrence patterns to environmental factors on the northeast continental shelf; developing ensemble SDMs for comparison with NCCOS models

Jun–Aug 2022 **NOAA NERTO Fellow**
National Centers for Coastal Ocean Sciences · Beaufort, NC (Remote)
Creating 2D orthoprojections of benthic coral reef plots in the Florida Keys National Marine Sanctuary; analyzed coral cover using Viscore and the Virtual Point Intercept (VPI) tool

2019–2021 **Environmental Scientist**
Environmental Logic, LLC · Lawrenceville, NJ
Groundwater sampling for contaminant testing; laboratory protocol development for NJDEP compliance

2017–2018 **Visiting Scholar**

Scripps Institution of Oceanography · La Jolla, CA

Analyzed benthic composition and coral population structure along wave energy and habitat structural complexity gradients at Rapa Nui; trained in large-area imaging and spatial point pattern analysis in the lab of Dr. Stuart Sandin

2014–2017 **Research Associate**

Millennium Nucleus for Ecology and Sustainable Management of Oceanic Islands · Coquimbo, Chile

Studied benthic composition and coral population structure along wave energy gradients at Rapa Nui

2010–2013 **Biologist**

US Geological Survey · St. John, USVI

Examined disease prevalence in *Acropora palmata*; discovered natural coral refugia from thermal stress in mangrove-lined bays

2004–2009 **Research Associate**

University of Miami, Rosenstiel School · Miami, FL

Coral restoration of endangered species; studied growth rates and survivorship in response to environmental stressors

Teaching Experience

Spring 2026 **Lecturer**

Brooklyn College, CUNY

BIOL 9000: Waves and Water Quality — Researching the Water Quality of the Hudson River (Lecture/Lab)

2023–2026 **Adjunct Lecturer**

Baruch College, CUNY

ENV 1003/1004 Fundamentals of Ecology (Lab/Recitation)

2023–2024 **Adjunct Lecturer**

Fashion Institute of Technology, SUNY

SC 211 Earth Science; SC 253 Ecology and Environmental Problems

2021–2023 **Adjunct Lecturer**

College of Staten Island, CUNY

BIO 272 Biostatistics; BIO 338 Animal Behavior; BIO 239 Ecology

2023 **Mentor**

City College of New York, CUNY

Data Science and Engineering Master Program — Capstone Project

2018 **Tutor**

Ivy Tutors Network

Coral Reefs: Biology, Geology, and Politics — Oberlin College course

2017 **Substitute Teacher**

Gifft Hill School · St. John, USVI

Grades 6–12

Publications

Peer-Reviewed Journal Articles

2025 Vörösmarty CJ, Trujillo M, Casaccia P, Cak A, Gardner KH, Greenfield DI, Groffman PM, Reinmann A, Solecki W, Serreze M, Waldman J, Bhatnagar JM, Brumberg H, Carnaval A, Cullmann J, Egendorf SP, Hauber ME, **Herlan J**, Kavouras I, Mason CE, Marcotullio P, McCracken M, Armstrong McKay D, Miszlivetz F, Muth T, Nomura Y, Penick CA, Rising J, Toledo-Crow R, Zarnoch C. A Climate BioStress Sentinel System: Identifying climate impacts from the genome to the urbanized biosphere. *Cell Reports Sustainability* 2: 100558. Perspective.

2022 Morales N, Coghlan A, Easton E, Friedlander A, **Herlan J**, Gaymer C. Recent first sightings of *Seriola rivoliana* at Rapa Nui, the result of range expansion or increased scientific effort? *J. Fish Biol.* 1–8. Short Communication.

2018 Easton E, Gaymer C, Friedlander A, **Herlan J**. Effects of herbivores, wave exposure, and depth on benthic coral communities of the Easter Island Ecoregion. *Marine and Freshwater Research* 69: 997–1006. Short Communication.

2014 Yates K, Rogers CS, **Herlan J**, Brooks G, Smiley N, Larson R. Diverse coral communities in mangrove habitats suggest a novel refuge from climate change. *Biogeosciences* 11: 4321–4337.

2014 Cooper W, Lirman D, Porter VanGroningen M, Parkinson J, **Herlan J**, McManus J. Assessing techniques to enhance early post-settlement survival of corals in situ for reef restoration. *Bulletin of Marine Science* 90(2): 651–664.

2010 Lirman D, Thyberg T, **Herlan J**, Hill C, Young-Lahiff C, Schopmeyer S, Huntington B. Propagation of the threatened staghorn coral *Acropora cervicornis*: Methods to minimize the impacts of fragment collection and maximize production. *Coral Reefs* 29(2): 729–735.

2008 Lirman D, DeAngelo G, Serafy J, Hazra A, Smith Hazra D, **Herlan J**, Luo J, Bellmund S, Wang J, Clausing R. Seasonal changes in the abundance and distribution of submerged aquatic vegetation in a highly managed coastal lagoon. *Hydrobiologia* 596: 105–120.

Manuscripts *In press*

In press **Herlan J**, McTague SA, Staniczenko PA, Loh JM, Santos R, James WR, Gleason A, Edwards CB, Gosnell JS, McDonald K, Friedlander AM, Gaymer CF. Spatial ecology of coral reefs at Rapa Nui (Easter Island): biotic mechanisms and environmental drivers of coral composition and dispersion along a wave exposure gradient. *Marine Ecology Progress Series.*

Manuscripts *In revision*

Accepted with minor revisions **Herlan J**, Staniczenko PA, Smith TB, McDonald K, Gosnell JS, Manne L, Friedlander AM, Gaymer CF. First quantitative assessment of coral bleaching at Rapa Nui (Easter Island) during the 2014–2017 global thermal event reveals predominantly mild stress responses. *Marine Biology* (revised manuscript submitted on August 19, 2026).

Manuscripts *In prep*

In prep **Herlan J**, Raval M, Loh JM. Dead *Pocillopora* skeletons shift from segregated to spatially mixed patches with live colonies along a wave-exposure gradient at Rapa Nui.

In prep **Herlan J**, Poti M, Winship A, Vasslides J. Mapping habitat for *Desmophyllum dianthus* on the Northwest Atlantic continental margin: a parsimonious species distribution model for marine spatial planning.

In prep **Herlan J**, Staniczenko PA, Smith TB, Brandtneris V, McDonald K, Gosnell JS, Edwards CB, Friedlander AM, Gaymer C. Wave exposure drives benthic habitat assembly and *Porites lobata* morphological plasticity at Rapa Nui.

In prep **Herlan J**, Rogers C. Effects of seawater temperature and storms on disease and survival of the threatened elkhorn coral *Acropora palmata* at three reef sites in St. John, US Virgin Islands (2010–2012)

In prep **Herlan J**, Rogers C. Spatially explicit drivers of coral bleaching and recovery in a coexisting mangrove–coral habitat during the 2010 seawater thermal event at Otter Creek, Hurricane Hole, USVI

Book Chapters

2016 Alvarado J, Aburto-Oropeza O, Abad R, Barraza E, Brandt M, Cantera J, Estrada P, Gaymer C, Guzman A, **Herlan J**, Mate J. Coral reef conservation in the Eastern Tropical Pacific. In: Glynn P, Manzello D, Enochs I (eds) *Coral Reefs of the Eastern Tropical Pacific: Persistence and Loss in a Dynamic Environment*. Springer.

Scientific Reports

2008 Bouchon C, Portillo P, Bouchon-Navaro Y, Louis M, Hoetjes P, De Meyer K, Macrae D, Armstrong H, Datadin V, Harding S, Mallela J, Parkinson R, Van Bochove J-W, Lirman D, **Herlan J**, Baker A, Collado L, Isaac SC. Status of Coral Reef Resources of the Lesser Antilles. In: Wilkinson C (ed) *Status of Coral Reefs of the World: 2008*. Australian Institute of Marine Science. pp. 265–280.

Conference Proceedings

2012 Rogers C, **Herlan J**. Life on the edge: corals in mangroves and climate change. *Proceedings 12th International Coral Reef Symposium*. Cairns, Australia.

2008 **Herlan J**, Lirman D. The development of a coral reef nursery program for the threatened coral *Acropora cervicornis* in Florida. *Proceedings 11th International Coral Reef Symposium*. Ft. Lauderdale, FL. Session 24–25.

2008 Lirman D, **Herlan J**, Capo T. Exposure to nutrients mitigates the negative impacts of sedimentation on coral growth in the Caribbean corals *Porites astreoides* and *Siderastrea siderea*. *Proceedings 11th International Coral Reef Symposium*. Ft. Lauderdale, FL. Session 19.

2008 Cooper W, Lirman D, Porter M, **Herlan J**, Parkinson J. Assessing the efficacy of in situ coral larval seeding under high rates of early post-settlement mortality. *Proceedings 11th International Coral Reef Symposium*. Ft. Lauderdale, FL. Session 24.

Conference Presentations

2026 **AGU Annual Meeting — Poster**

San Francisco, CA

Spatial ecology of coral reefs at Rapa Nui (Easter Island): biotic mechanisms and environmental drivers of coral composition and dispersion along a wave exposure gradient (Abstract ID: 2094456, Session: OS21C Advances in Nearshore Processes IV)

2026 **Student Conference on Conservation Science — Poster**

American Museum of Natural History, NY

Dispersion and neighborhood co-occurrence of *Pocillopora* corals at Rapa Nui

2026 **12th Biennial Education and Science Forum — Poster**

City College of New York, CUNY

Mapping habitat for *Desmophyllum dianthus* on the Northwest Atlantic continental margin: a parsimonious species distribution model for marine spatial planning—**2nd Place: Best Poster Presentation Award | Living Marine Resources and the Blue Economy**

2026 **Temperate Coral Research Conference — Talk**

Southern Connecticut State University, CT

Spatial Signatures Across a Wave-Exposure Gradient: Exploring a Transferable Analytical Framework that May Inform the Ecology of *Astangia poculata* of shallow coastal habitats in the Mid-Atlantic and New England regions of the US.

2026 **AGU Ocean Sciences Meeting — Talk**

Glasgow, Scotland

Impact of wave exposure and habitat structural complexity on the reefscape ecology at Rapa Nui (Abstract ID: ME31A-05). *Abstract withdrawn due to a family emergency.*

2025 **11th Biennial Education and Science Forum — Talk**

University of Maryland Eastern Shore

Impact of wave exposure on the abundance and dispersion of marine benthic organisms at Rapa Nui

2024 **AGU Annual Meeting — Talk**

Washington, DC

Impact of wave exposure on the abundance and dispersion of marine benthic organisms at Rapa Nui (Abstract ID: 1762322, Paper No. EP31B-04)

2024 **ESA Annual Meeting — Talk**

Long Beach, CA

Impact of wave exposure on coral dispersion at Rapa Nui (Abstract ID: 67606)

2024 **Student Conference on Conservation Science — Talk**

American Museum of Natural History, NY

Dispersion of coral along a wave exposure gradient at Rapa Nui

2024 **Mid-Atlantic Chapter ESA Annual Conference — Talk**

Kutztown University, PA

Dispersion of coral along a wave exposure gradient at Rapa Nui

2023 **Climate Stress across the Biosphere — Talk**

Advanced Science Research Center, CUNY

Coral Reefs in the Anthropocene

2023 **CUNY 16th Annual Research Symposium — Poster**

City College of New York

Spatial ecology of pocilloporid coral at Rapa Nui

2022 **10th Biennial Education and Science Forum — Talk**

Florida A&M University

The influence of wave exposure on the functional morphologies of coral at American Samoa

2021 **14th International Coral Reef Symposium — Poster (remote)**

Bremen, Germany

Population variation of *Porites lobata* and *Pocillopora* spp. along wave-energy gradients at Rapa Nui (Abstract ID: 2913)

2020 **3rd MIT Quantitative Ecology Meeting — Poster**
Massachusetts Institute of Technology
Spatial dispersion of the coral *Pocillopora* spp. along a wave-energy gradient at Rapa Nui

2016 **13th International Coral Reef Symposium — Poster**
Honolulu, HI
Coral bleaching in Rapa Nui: relation to the El Niño Southern Oscillation? (Abstract ID: 29956, Poster No. 319)

2015 **XXXV Congreso Ciencias del Mar — Poster**
Coquimbo, Chile
Blanqueamiento de corales en Rapa Nui. Utilizando un modelo de predicción

2013 **Coastal and Estuarine Research Federation (CERF) — Poster**
San Diego, CA
Shading may reduce bleaching of corals in a mangrove ecosystem (Abstract ID: 1746850)

2008 **11th International Coral Reef Symposium — Talk**
Ft. Lauderdale, FL
Development of a coral reef nursery program for the threatened coral *Acropora cervicornis* in Florida (Abstract 25-25: Oral Mini-Symposium 24: Reef Restoration)

Grants and Funding

2026 Hudson River Park Trust Research Grant ($20,000) — Co-investigator

2024–2025 Hudson River Park CUNY Visiting Scholars Award ($10,000) — Co-investigator

2024–2025 PSC-CUNY Research Award Cycle 55 ($5,936) — Co-investigator

2022 USGS/CCNY Internship Program ($4,000)

2015 Crowdfunding — experiment.com ($1,025) for field equipment

2015 Idea Wild ($735) for field equipment

Fellowships and Scholarships

2023–Present NOAA-EPP/MSI CESSRST-II Fellow (Grant NA22SEC4810016)

2020–2022 NOAA-EPP/MSI CESSRST Fellow (Grant NA16SEC4810008)

2022–2023 CUNY Science Scholarship ($18,031 stipend)

2022–2023 Graduate Assistantship B ($12,956 stipend)

Professional Service and Outreach

2026 SCCS-NY 2026 Workshops, *From Photos to Spatial Pattern: Large-Area Imaging for Coral Reef Conservation*, American Museum of Natural History—New York, NY

2026 Oral Presentation, New York Species Distribution Modeling Discussion Group (NY-SDM), American Museum of Natural History—New York, NY

2026 Oral Presentation, 2026 Estuarine Sanctuary Management Plan (ESMP): Sanctuary Science Meeting, Hudson River Park Trust

2024 Lab research stall, Release of the Fishes, Pier 40 at Hudson River Park

2024 Ask-a-Scientist Speaker, Pier 57 at Hudson River Park

Peer Review	*Coral Reefs* — Journal of the International Society for Reef Studies
Memberships	International Coral Reef Society; Ecological Society of America; Association for the Sciences of Limnology and Oceanography; Society for Industrial and Applied Mathematics

Technical Skills

Field	SCUBA diving (AAUS and NPS blue card scientific diver candidate), Motorboat Operator Certification Course (MOCC), coral reef surveying, large-area imaging, Structure-from-Motion photogrammetry, water quality monitoring
Laboratory	Coral husbandry, water quality analysis, microscopy (basic knowledge), specimen preparation (basic knowledge)
Computing	R, Python, GIS (ArcGIS Pro), Agisoft Metashape, Viscore, TagLab, CoralNet, CoralNet-Toolbox, Adobe Creative Suite
Statistical	Generalized linear, additive, and mixed models, spatial point pattern analysis, bootstrapping, time series analysis, Random Forest, Extra Trees, Gradient Boosting, XGB
Languages	English (native), Spanish (working proficiency)

References

Phillip P.A. Staniczenko
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J. Stephen Gosnell
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Kyle McDonald
Terry Elkes Professor, Earth and Atmospheric Sciences, City College of New York, CUNY
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